

Token Activation Map to Visually Explain Multimodal LLMs

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Abstract

Multimodal large language models (MLLMs) are broadly empowering various fields. Despite their advancements, the explainability of MLLMs remains less explored, hindering deeper understanding, model credibility, and effective visualization. Unlike conventional vision models (e.g., CNNs, ViTs, CLIP) that produce a single output, MLLMs generate sequences of tokens progressively, where each generated token depends on the previous context. Therefore, earlier context tokens can introduce redundant activations that interfere with the explanation of later tokens beyond their original information. Existing studies often overlook this issue, but our observations reveal that these redundant correlations can significantly hurt the reliability of explanations. To address this, we propose an estimated causal inference method to mitigate the interference of context to achieve high-quality MLLM explanation, with a novel rank Gaussian filter to further reduce activation noises. We term this method Token Activation Map (TAM) to highlight the consideration of interactions between tokens. TAM also indicates that it excels at explaining multiple tokens of MLLM, which is different from the Class Activation Map (CAM) for a single prediction. Our TAM method significantly outperforms existing SoTA methods, showcasing high-quality visualization results that can be utilized for various scenarios, such as object localization, failure case analysis, video visualization, MLLMs visual comparison, and model understanding (e.g., color, shape, action, location, visual reasoning, multi-turn conversation, etc). The code is available at github.com/xmed-lab/TAM.

1. Introduction

Multimodal large language models (MLLMs or Multimodal LLMs) increasingly empower wide applications [17, 29, 30, 34], enabling multimodal inputs (e.g., images, videos, text) and human-like conversations. Although extensive efforts have been devoted to enhancing the performance of

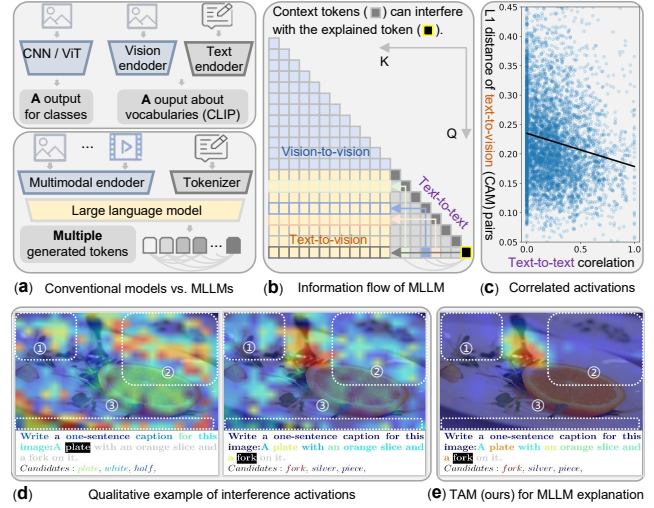


Figure 1. Illustration of Motivation. (a) MLLMs generate multiple tokens beyond a single output. (b) The information flow indicates that MLLM generates tokens progressively, where one generated token (each row) is correlated with the context (prompt + earlier answer tokens). (c) We randomly pair CAMs and count their L1 distance against text correlation. Higher text correlation corresponds to lower distance, indicating concurrent interferences. (d) The example shows that the context token “plate” introduces interference activations (marked by white boxes) to the later explained token “fork”. (e) Our TAM reveals the original information of the explained token “fork”, eliminating context interferences and beyond CAM for a single output in (d). Results in (c, d, e) derive from Qwen2-VL-2B on the COCO Caption dataset.

MLLMs [4, 16, 35, 47, 51], research on MLLM explainability remains less explored. It is crucial for user trust, model understanding, case analysis, and visualization. However, exploring the explainability of MLLMs is more challenging than conventional visual models (such as CNN [24], ViT [23], CLIP [41]), where MLLMs progressively generate multiple tokens beyond a single output as Fig. 1a. Thus, we aim to explain all tokens of MLLM, revealing the visual cues via token-level activation maps (e.g., Fig. 1e).

Current MLLM explanation works [7, 48, 55] are generally borrowed from techniques [38, 45] for conventional models. For example, LLaVA-CAM [55] uses Smooth

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Grad-CAM++ [38] to explain LLaVA [35]. The self-attention weights [49] are applied to understand math problem-solving [48]. Besides, LVLM-Interpret [7] integrates some conventional methods as a tool. Unfortunately, existing visual explainability methods are designed for conventional models [23, 24, 41] that produce a single output, such as class activation maps (CAM) series [9, 26, 45, 56], attention-based approaches [1, 28], attention with relevance [2, 11], model-agnostic methods [36, 43], or multimodal explanation [3, 19, 31, 32]. *However, these methods overlook that MLLMs generate multiple tokens progressively, where earlier context tokens interfere with later tokens to be explained.* As shown in Fig. 1d, the conventional CAM [56] for the token “fork” is confounded by numerous redundant activations concurrent with the token “plate” in context (context refers to prompt and earlier answer tokens than the explained token). Quantitative analysis in Fig. 1c reveals that this issue is widespread, where correlated tokens lead to close and interference activations. These interferences obscure the original information conveyed by the explained token, significantly degrading the explainability of MLLMs. Notably, this activation interference across tokens is unique to MLLMs and has been overlooked.

We aim to reveal the original information for the explained token while minimizing interference from context, as discussed above. Herein, the original information reveals a causal relation [40] between the current token and the input, while the interferences represent the correlations that need to be excluded. Inspired by the potential outcome model (POM) of causal inference [27], we propose the estimated causal inference to explore this causal relation. Unlike raw causal inference by comparing ready-made results (yes vs. no), there is no single given output to represent the “no” result. Therefore, we estimated interferences from multiple context tokens that are highly related to the target, with a scale factor optimized by the least-squares method to generate the refined causal activation map. We also propose a novel rank Gaussian filter to reduce noises, thereby further enhancing the quality of activation maps. The above methods are collectively referred to as token activation map (TAM), emphasizing the consideration of interactions between tokens. TAM also indicates that it excels at explaining tokens in multiple rounds of MLLM, which differs from the Class Activation Map (CAM) for a single prediction.

Experimentally, the proposed TAM demonstrates quantitative improvement over existing SoTA explainability methods by more than 8.96% on the COCO Caption dataset [13] and 8.54% on the OpenPSG dataset [57] when applied to visually explain the Qwen2-VL-2B [51]. Meanwhile, it is complementary to existing methods to boost their performance. We further verify the scalability and applicability on 7 advanced MLLMs from the series of Qwen2-VL [51], InternVL2.5 [15] and LLaVA1.5 [35], where the

performance improvements range from 5.45% to 11.0% on COCO Caption dataset [13]. Besides quantitative results, we conduct qualitative experiments to understand MLLM attributes about color, shape, action, location, etc. Furthermore, we illustrate that TAM is capable of explaining failure cases on the QK-VQA dataset. [37], and it supports video visualization well on the STAR dataset [52]. In addition, it can be used as a tool to visually compare MLLMs qualitatively. Moreover, TAM shows wide applicability to explain multi-turn conversations, multi-image scenarios, and visual reasoning (see the catalog Table 5 in supplementary for extensive examples). Our work has three main contributions:

- We introduce TAM, a novel approach to explain MLLMs, which produces high-quality activation maps for multiple generative tokens by incorporating the estimated causal inference to alleviate interference from context tokens, with the rank Gaussian filter to reduce activation noises.
- TAM significantly outperforms current methods for MLLM explanation, while also complementing them.
- TAM demonstrates wide applicability and scalability, serving as a versatile tool for object localization, model understanding, failure case analysis, video visualization, MLLM comparison, and supporting diverse scenarios (e.g., multi-turn, multi-image, visual reasoning).

2. Related Work

Visual Explainability for Multimodal LLM. The emergence of MLLMs such as LLaVA [35], Qwen2-VL [51], InternVL [15], and closed-source GPT-4o [25] has significantly impacted various tasks involving multimodal inputs like text, images, and video. We mainly focus on the visual explanation of MLLMs via the activation maps for the input image or video frames. Regarding the modality perspective, the explainability of MLLMs is partially related to methods to explain multimodal models such as CLIP Surgery [32] and transformer register [19] designed for CLIP [41]; Interpret [3] and Bi-Modal [10] for BERT [20]. From the technical aspect, conventional explanation methods are also related, including class activation map series (CAM [56], Grad-CAM [45], Grad-CAM++ [9], LayerCAM [26]), attention-based mechanisms (LRP [28], Rollout [1]), the combination (AttnLRP [2], Grad×AttnRoll [11]), or model-agnostic methods (LIME [43], SHAP [36]).

Although existing methods may be valid to explain MLLMs, they are not the optimal solution. Because they are usually designed for conventional models (e.g., CNN [24], ViT [23], CLIP [41]) that produce a single output rather than generating multiple tokens progressively in MLLMs. We have observed that these earlier context tokens interfere with later tokens by introducing redundant visual activations, as shown in Fig. 1. This phenomenon is first studied by us, solved by a novel estimated causal inference method, which is specially designed for MLLM. Notably,

some workshop papers (LLaVA-CAM [55] and LVL-Interpret [7]) applied conventional methods [38, 44] to explain LLaVA [35], but they also overlooked the interference from context tokens and treated each token independently.

Causal Model. Causal inference [54] seeks to establish causal relation [40] between variables, focusing on how changes in one variable influence another. In contrast to statistical correlation analysis, it emphasizes causal relations. Another related concept, causal intervention [27], involves manipulating variables using a do operator. Model-agnostic explainability methods, such as LIME [43] and SHAP [36], can be viewed as causal interventions that explore the relationship between inputs and structured outputs by masking inputs or selecting subsets [12]. This concept is also applied to transformers through token masking (CLEANN [44]) or hidden state replacement [39]. However, causal intervention for MLLMs is impractical. Because there are too many input tokens are outputs in MLLM, requiring unbearable inference times. Besides, modified input leads to a changed context every inference, which is hard to evaluate. Our method is inspired by the potential outcome model (POM) to achieve causal inference without additional model inferences. Our contribution lies in estimating the unprocessed output from multiple tokens rather than a single ready-made output, considering the uniqueness of MLLMs.

Transformer Denoising. Activation maps of transformers [49] usually present many noise activations, impeding high-quality visualization. Recent studies give their explanations about it, including attention sink [53], lack of register [19], and redundant features across classes [32]. In MLLM, these noises still exist obversely, even system tokens already play the role of registers, or deploying the feature surgery to mitigate redundant features (See Table 2). To solve this problem practically, we aim to introduce denoising filters to mitigate it as post-processing. Since these noises belong to salt-and-pepper noise morphologically, we apply the median filter, adaptive median filter [8], and the Gaussian filter. However, these filters are not the optimal solution, where the Gaussian filter keeps too much noise signal and the median filter overlooks smaller responses. Therefore, we propose the rank Gaussian filter, a novel, simple, and effective filter to denoise transformer activation. It merges ranked values within a sliding window, weighted summed by a 1-d Gaussian kernel with the center at the median rank. Besides, we improve the Gaussian kernel by the coefficient of variation for more robust results. Overall, our method addresses the activation noises in a practical aspect through a new filter.

3. Method

In this section, we first provide an overview of the proposed Token Activation Map (TAM), as illustrated in Fig. 2. Next, we elaborate on two key modules within TAM: the estimated causal inference and the rank Gaussian filter. Finally,

we introduce three metrics designed to enable fine-grained evaluations of the explanations for MLLMs.

3.1. Token Activation Map

TAM aims to depict explainable activation maps for multiple tokens generated from MLLMs by leveraging causal inference to mitigate inter-token interference. To be more specific, given the visual data and prompt text as the input, the MLLM is employed to progressively generate the visual features $\mathbf{F}^v \in \mathbb{R}^{n_v \times c}$, prompt features $\mathbf{F}^p \in \mathbb{R}^{n_p \times c}$ and answer features $\mathbf{F}^a \in \mathbb{R}^{n_a \times c}$. n_v , n_p , n_a and c indicate the number of three type of features, and the feature dimension, respectively. Next, a token classifier, instantiated as a fully connected layer, is used to generate answer tokens $\mathbf{t}^a = \{t_1^a \dots t_{n_a}^a\}$ from these features. Prompt tokens are defined as $\mathbf{t}^p = \{t_1^p \dots t_{n_p}^p\}$ similarly. Based on the variables described above, the visual activation maps for the prompt tokens and answer tokens are calculated as follows:

$$\begin{aligned} \mathbf{A}_i^p &= \lfloor \mathbf{F}^v \mathbf{w}_{t_i^p} \rfloor_+, i \in [1, n_p] \\ \mathbf{A}_i^a &= \lfloor \mathbf{F}^v \mathbf{w}_{t_i^a} \rfloor_+, i \in [1, n_a], \end{aligned} \quad (1)$$

where $\mathbf{A}_i^p \in \mathbb{R}^{n_v \times 1}$ and $\mathbf{A}_i^a \in \mathbb{R}^{n_v \times 1}$ present the activation map for the i -th prompt token t_i^p and answer token t_i^a . $\mathbf{w}_{t_i^p} \in \mathbb{R}^{c \times 1}$ and $\mathbf{w}_{t_i^a} \in \mathbb{R}^{c \times 1}$ are the corresponding weight vectors within the token classifier. $\lfloor \cdot \rfloor_+$ is the function to keep positive activations.

Then, the activation maps for prompt and answer tokens are concatenated ($\perp$) as $\mathbf{A}_{:n_p+n_a} = \mathbf{A}^p \perp \mathbf{A}^a$. For the i -th answer token, its context token is formulated as $\mathbf{A}_{:n_p+i-1}$, including all earlier context maps (prompt tokens + answer tokens earlier than i). On top of it, the refined activation maps for answer tokens are formulated as follows:

$$\bar{\mathbf{A}}_i^a = \lfloor \mathcal{D}(\mathbf{A}_i^a - s\mathcal{E}(\mathbf{A}_{:n_p+i-1})) \rfloor_+, i \in [1, n_a], \quad (2)$$

where $\bar{\mathbf{A}}_i^a$ indicates the refined activation map for the i -th answer token. The number of i increases sequentially from 1 to n_a . $\mathbf{A}_i^a - s\mathcal{E}(\mathbf{A}_{:n_p+i-1})$ is the proposed estimated causal inference that produces the causal activation map with the estimation function $\mathcal{E}$ described in Sec. 3.2. $\mathcal{D}$ is the proposed rank Gaussian filter module in Sec. 3.3.

In the last, we concatenate ($\perp$) the refined visual activation map $\bar{\mathbf{A}}_i^a$ of the i -th answer token with the raw textual relevance $\mathbf{r}_i \in \mathbb{R}^{n_p+i-1}$ for a multimodal activation map (image or video with text) $\mathbf{M}_i \in \mathbb{R}^{n_v+n_p+i-1}$ as follows:

$$\mathbf{M}_i = \mathcal{N}(\bar{\mathbf{A}}_i^a \perp \mathbf{r}_i), i \in [1, n_a], \quad (3)$$

where $\mathcal{N}()$ is a min-max normalization function for visualizing visual and textual activations at an aligned level.

3.2. Estimated Causal Inference

Overall, Eq. 2 optimizes later visual activation maps based on visual activation maps of all earlier context tokens. To

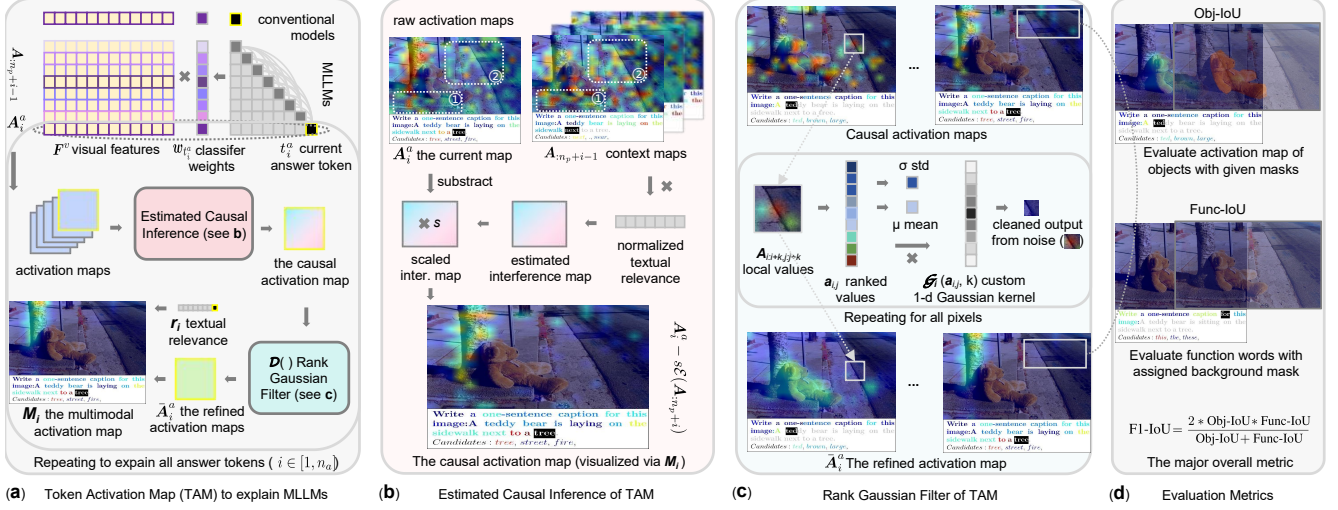


Figure 2. Illustration of method. (a, Eq. 1 - Eq. 3) The overall framework of TAM. (b, Eq. 4 - Eq. 5) Details of the estimated causal inference module. (c, Eq. 6 - Eq. 7) Details of the rank Gaussian filter module. (d, Eq. 8 - Eq. 10) Fine-grained evaluation metrics.

be more specific, causal inference is employed to investigate the correct causal relation between the current answer token and visual tokens, eliminating interferences from prompt tokens and preceding answer tokens. To this end, we quantify the interference map from the context tokens for the i -th answer token as follows:

$$\mathcal{E}(A_{:n_p+i-1}) = \sum_{k=1}^{n_p+i-1} \frac{r_i^k}{\sum \mathbf{r}_i + \epsilon} A_k \quad (4)$$

$s.t. r_i^k = 0, \text{ if } t_i^a = t_k,$

where r_i^k is the textual relevance between the k -th context token t_k and the current answer token t_i^a . The relevance is obtained from activations among textual tokens $[(F^p \perp F^a)w_{t_i^a}]_+$ following Eq. 1. If the current context token t_k is the same as the answer token t_i^a , we set their relevance as 0, *i.e.*, $r_i^k = 0$, to avoid eliminating visual activations for tokens identical to the current answer token.

Overall, Eq. 4 aims for each answer token to encapsulate more visual information, which implies minimizing similarity to the visual semantics of previously context tokens. Meanwhile, we use the scale factor s in Eq. 2 to control the scale of the interference map, ensuring that the influence of the interference map is removed from the activation map at a comparable degree. Specifically, we employ the least squares method to minimize the residual between the two maps, for the optimal value of s , as follows:

$$s = \arg \min_s \sum_{j=1}^{n_v} (A_{j,i}^a - s\mathcal{E}(A_{j:n_p+i-1}))^2. \quad (5)$$

After processing by the estimated causal inference module, there are still many salt-and-pepper noises in activation

maps. To address this issue, we will elaborate on the solution in the following section.

3.3. Rank Gaussian Filter

As described in Sec. 2 the activation noise exists obviously and belongs to the salt-and-pepper type. Different from existing methods [19, 32] with suboptimal performances on MLLM, we address this problem in a new aspect: image denoising. Compared with existing filters like the Gaussian filter that keeps too much noise signal and the median filter which overlooks smaller responses, our rank Gaussian filter is more robust beyond existing methods. The overall equation of the rank Gaussian filter for any activation map $A_{i,j} \in \mathbb{R}^{h \times w}$ is written as:

$$\mathcal{D}(A_{i,j}, k) = \sum A_{i,j} \mathcal{G}(\mathbf{a}_{i,j}, k), \forall i \in H, \forall j \in W \quad (6)$$

$s.t. \mathbf{a}_{i,j} = \text{rank}(A_{i:i+k, j:j+k}).$

Herein, $A_{i:i+k, j:j+k}$ indicates local values within a sliding window at kernel size k . We sort them to get a ranked array to multiply with a custom 1-d Gaussian kernel $\mathcal{G}()$. This custom kernel allows the median to have the largest weight, meanwhile aggregating signals ranked close to the median. It can be regarded as a kind of smoothed median filter to enhance the robustness weighted by the Gaussian kernel.

For the custom 1-d Gaussian kernel, we formulate it as Eq. 7. Specifically, $(i - k^2/2)^2$ indicates the distance from the median rank ($k^2/2$, size exact division 2), and $\sigma(\mathbf{a})/\mu(\mathbf{a})$ is the coefficient of variation which is more stable than the standard division $\sigma()$ in general Gaussian kernel using the mean $\mu()$ only.

$$\mathcal{G}_i(\mathbf{a}, k) = e^{-\frac{(i - k^2/2)^2}{2(\sigma(\mathbf{a})/\mu(\mathbf{a}))^2}}, \forall i \in k^2 \quad (7)$$

3.4. Evaluation for MLLM Explanation

Unlike conventional models with a single output, MLLMs generate multiple tokens. To evaluate how well activation maps correspond to objects, we propose two fine-grained IoU-based metrics: Obj-IoU for objects with manual masks, and Func-IoU for backgrounds. These metrics focus on plausibility instead of faithfulness (see concept differences in Supp. E). Plausibility better suits MLLMs since perturbation-based faithfulness tests [2, 11] which cause inconsistent text generation, making evaluation invalid (see detailed reasons in Supp. E).

To be specific, we formulate Obj-IoU in Eq. 8, counting the IoU between activation map ($\mathbf{A}_i$) after binarization ($\mathcal{B}()$ using opencv OTSU auto threshold) and the ground-truth mask $\mathbf{G}_i$. Since the object classes may be unfixed, we count the IoU for all tokens at size o over the dataset without an average on classes. We obtain the mask by matching the token to mask names. For a word or phrase containing multiple tokens, we record their max IoU for a single object.

$$\text{Obj-IoU} = \frac{1}{o} \sum_{i=1}^o \frac{\mathcal{B}(\mathbf{A}_i) \cap \mathbf{G}_i}{\mathcal{B}(\mathbf{A}_i) \cup \mathbf{G}_i} \quad (8)$$

Func-IoU measures the activation degree of tokens that are unreadable from the image (e.g., “is”, “and”, “so”, “the”). Higher Func-IoU indicates fewer false positives and unnecessary responses. Specifically, we record the mean binarization thresholds b_i (opencv OTSU auto threshold) for all noun tokens in the same image to divide foreground and background on function words. Activation map lower than it ($\mathbf{A}_i < b_i$) is the background prediction used to count IoU with the all-one matrix $\mathbf{J}$ (ground-truth is all background class). It is applied to total u tokens over the whole dataset.

$$\text{Func-IoU} = \frac{1}{u} \sum_{i=1}^u \frac{(\mathbf{A}_i < b_i) \cap \mathbf{J}}{(\mathbf{A}_i < b_i) \cup \mathbf{J}} \quad (9)$$

We try to merge the above IoUs as the overall major metrics. We find their average is insufficient to measure biased and low-quality activation maps. For example, if one method predicts almost all tokens to be background, the Obj-IoU is very low but Func-IoU is high, forming a certain inverse ratio (e.g., 5.74% vs. 96.5% of Attention-Rollout [1] in Table 3). One suitable solution is to use the F1 value to merge metrics with a certain inverse ratio. Thus, we have the F1-IoU to evaluate the overall plausibility stably:

$$\text{F1-IoU} = \frac{2 * \text{Obj-IoU} * \text{Func-IoU}}{\text{Obj-IoU} + \text{Func-IoU}} \quad (10)$$

4. Experiments

4.1. Setup

Datasets. For quantitative experiments, we use datasets with texts and pixel-level annotations. Our main dataset is the COCO Caption dataset [13], which draws images and

masks from COCO2014 [33]. Since explainability methods require testing only, we use its 5K-image minival split without training. Additional datasets include GrandF [42] (1K images) and OpenPSG [57] (3,176-image validation set). Masks in COCO Caption and GrandF are manually annotated, while OpenPSG masks are integrated by Rasheed et al. [42]. For datasets without masks, we perform qualitative visualization tests (Sec. 4.3), including attribute-only images and the QK-VQA [37] validation set (5,046 VQA samples with textual answers). We also qualitatively evaluate the STAR video dataset [52] (914 videos).

Implementations. We use the SciPy minimize function with the BFGS method to optimize the scale factor s in Eq. 5. The only hyperparameter, the kernel size k in Eq. 7, is set to 3, as our observations indicate that salt-and-pepper noises typically cluster between 1 to 5 tokens. A kernel size of 3, consisting of 9 elements, is deemed suitable. Eq. 9 incorporates part-of-speech identification. For Func-IoU and Obj-IoU, we tag each word’s part of speech using the pos_tag function from the NLTK Python package (version 3.8.1, see tag details in Supp. E). For word matching to mask names, we employ the lemmatize function from nltk.stem. Note that the thresholds in Eq. 8 and Eq. 9 are automatically derived from OpenCV’s OTSU method, requiring no manual operation. Details of the baselines are provided in Supp. D.

4.2. Quantitative Results

ECI	RGF	Obj-IoU (%)	Func-IoU (%)	F1-IoU (%)
$\times$	$\times$	21.23	51.93	30.14
$\checkmark$	$\times$	22.41	69.03	33.84 $\pm$ 3.7
$\times$	$\checkmark$	24.82	43.34	31.57 $\pm$ 1.33
$\checkmark$	$\checkmark$	27.37	68.44	39.1 $\pm$ 8.96

Table 1. Ablation study on the COCO Caption [13] dataset using Qwen2-VL-2B [51]. These two modules are mutually beneficial, the combination exceeds the sum of their individual gains. ECI indicates the proposed estimated causal inference, and RGF is the rank Gaussian filter. Metrics are IoU for object words, IoU for function words, and their F1-score-like combination, respectively.

Setting	Obj-IoU (%)	Func-IoU (%)	F1-IoU (%)
Baseline	21.23	51.93	30.14
Replace rank Gaussian filter			
Adaptive Median [8]	25.48	68.2	37.1
Median Filter	26.01	68.26	37.67
Gaussian Filter	26.56	67.78	38.16
Replace estimated causal inference			
Feature surgery [32]	18.5	48.66	26.81
ECI-mean	27.84	49.85	35.72
ECI-attnWeights	27.11	54.61	36.23
Ours	27.37	68.44	39.1

Table 2. Effectiveness study on the COCO Caption [13] dataset using Qwen2-VL-2B [51]. ECI-mean and ECI-attnWeights are candidate implementations of ECI described in Supp. D.

Ablation and Effectiveness. We present the ablation study results in Table 1, which indicates that each module of the TAM is valuable. Specifically, the estimated causal infer-

Method	Type	COCO Caption			GrandF			OpenPSG		
		Obj-IoU	Func-IoU	F1-IoU	Obj-IoU	Func-IoU	F1-IoU	Obj-IoU	Func-IoU	F1-IoU
Grad-CAM [45]	Gradient	21.23	51.93	30.14	17.85	62.15	27.74	22.93	48.57	31.15
Grad-CAM++ [9]		19.52	62.83	29.78	17.3	73.42	28.01	22.21	59.95	32.41
Grad-Rollout [1]		1.27	99.51	2.51	1.4	99.61	2.77	1.57	99.58	13.08
Layer-CAM [26]	Attention	11.43	84.88	20.15	13.11	82.09	22.62	14.12	85.29	24.22
Attention		8.2	92.87	15.07	9.6	93.56	17.42	10.58	94.28	19.03
Attention-Rollout [1]		5.74	96.5	10.83	7.21	96.65	13.42	7.94	97.04	14.68
CP-LRP [5]	Combination	9.9	53.97	16.73	12.61	53.24	20.39	13.3	53.36	21.3
Attn-LRP [2]		9.92	52.41	16.69	12.15	52.19	19.72	12.78	52.26	20.54
CAM [56]		21.23	51.93	30.14	17.85	62.15	27.74	22.93	48.57	31.15
Archi.-Surgery [32]	Logit	15.69	63.82	25.19	16.59	62.28	26.2	19.83	58.77	29.65
TAM (ours)		27.37	68.44	39.1	18.65	88.97	30.83	26.26	92.99	40.95
Archi.-Surgery [32]	+ Ours	+4.13	+10.68	+6.12	+1.41	+26.48	+3.74	+1.59	+31.92	+5.0
Grad-CAM++ [9]		+2.83	+3.91	+3.7	+1.26	+5.39	+2.04	+2.75	+9.44	+4.3
Layer-CAM [26]		+5.04	-7.95	+6.98	+2.79	+0.52	+4.05	+6.62	-2.49	+8.95
CAM [56]		+6.14	+16.52	+8.96	+0.79	+26.82	+3.09	+3.33	+44.42	+9.8

Table 3. Comparison with SoTA methods using Qwen2-VL-2B on diverse datasets. We adopt the “Logit” type that uses the classifier weights in Eq. 1, without back-propagation in “Gradient” and “Combination” (Gradient + Attention). Besides, we support FlashAttention [18] and SdpaAttention, which do not return attention weights, while “Attention” and “Combination” rely on it. Our TAM is also complementary to existing methods with gains marked by “+”. Note, CAM and Grad-CAM are equivalent as discussed in Supp. D. The major metric is the F1-IoU (%) to reflect the overall result, merged from IoUs for Object words (Obj-IoU) and function words (Func-IoU).

CAM [56] +Ours	COCO Caption			GrandF			OpenPSG		
	Obj-IoU	Func-IoU	F1-IoU	Obj-IoU	Func-IoU	F1-IoU	Obj-IoU	Func-IoU	F1-IoU
LLaVA1.5-7B [35]	+4.47	+18.27	+7.97	+0.65	+11.67	+2.47	+3.47	+9.51	+5.16
LLaVA1.5-13B [35]	+4.3	+7.31	+5.45	+0.76	+7.03	+2.11	+4.22	+11.51	+6.37
InternVL2.5-2B [16]	+5.44	+19.48	+8.57	+2.2	+48.29	+8.47	+3.24	+40.45	+8.65
InternVL2.5-4B [16]	+3.53	+22.17	+7.14	+1.62	+45.19	+7.56	+2.2	+55.05	+10.85
InternVL2.5-8B [16]	+5.38	+2.13	+6.93	+3.53	+28.53	+7.02	+3.27	+26.54	+6.44
Qwen2-VL-2B [51]	+6.14	+16.52	+8.96	+0.79	+26.82	+3.09	+3.33	+44.49	+9.8
Qwen2-VL-7B [51]	+5.62	+29.4	+11.01	+1.29	+22.54	+3.4	+3.53	+46.94	+11.15
Mean Improvements	+4.98	+16.47	+8.01	+1.55	+27.15	+4.88	+3.32	+33.5	+8.35

Table 4. Explainability improvements of TAM on diverse MLLMs and datasets. Specific results are listed in Table 6 of Supp. K.

ence significantly improves Func-IoU by reducing context interference on function words, while the rank Gaussian filter performs better for object words. Importantly, these modules are mutually beneficial; their combination leads to an F1-IoU improvement of 8.96%, exceeding the total of their individual contributions (3.7% and 1.33%).

We further assess the effectiveness of these modules by comparing them with other methods, as shown in Table 2. Replacing our proposed rank Gaussian filter with alternatives yields lower performance, such as the adaptive median filter [8], the Gaussian filter, and the median filter. Although the differences are less pronounced than those seen with the causal inference module, denoising remains a challenging area. Our method outperforms feature surgery [32] (which operates on the class dimension) by 12.29% and also surpasses other ECI implementations in Supp. D.

Comparison with SoTA Methods. Table 3 presents a comparison between TAM and SoTA methods. Our TAM outperforms all others in the major F1-IoU and Obj-IoU metrics. While some methods, such as Grad-Rollout [1] and Attention, achieve high Func-IoU by producing a limited number of activations, their performance on other metrics re-

mains low. The second-best method is Grad-CAM++ [9] on the COCO Caption [13] and OpenPSG [57] datasets, while CAM [56] and Grad-CAM [45] rank second on GrandF [42]. Our method surpasses these approaches by 8.96%, 2.82%, and 8.54% on these datasets, respectively. Furthermore, TAM is complementary to existing methods and enhances them significantly, with maximum gains of 6.62%, 44.42%, and 9.8% across the three metrics, respectively.

Applicability and Scalability. We evaluate the applicability on three prominent MLLMs, as shown in Table 4, and assess its scalability across various model sizes. The results in Table 3 indicate that TAM consistently outperforms the baseline across all MLLMs and model sizes by significant margins. This demonstrates that TAM can be effectively applied to a range of MLLMs while providing superior explainability compared to existing methods. Moreover, TAM supports MLLMs comparison, complementing quantitative results. It introduces a new evaluation perspective by highlighting visual alignment quality, as illustrated in Fig. 8.

4.3. Qualitative Results

Comparison of Explainability Methods.



Figure 3. Visual comparison between TAM and SoTA methods on the COCO Caption dataset [13] using the Qwen2-VL-2B [51] model. The “is” in the last row is a functional word with background ground-truth. More examples are shown in Supp. G including complex cases.



Figure 4. TAM presents high-quality localization results on diverse datasets [13, 42, 57] under the Qwen2-VL-7B [51] model.

We perform comparisons between the proposed TAM and SoTA explanation methods, as illustrated in Fig. 3. The visualization results clearly show that TAM more accurately highlights target objects and renders fewer false activations on function words. In contrast, attention-based methods [1, 2, 5] produce only a limited number of activations, while gradient-based methods [9, 45] present too many correlated responses and noise, performing significantly worse than our approach. Additionally, we present more complex examples in Supp. G, where TAM notably outperforms existing SoTA methods. TAM excels at explaining multiple tokens of MLLMs. In Supp. C, results show how TAM focuses on important tokens, whereas the baseline methods generate many redundant activations.

High-quality Localization Results. We visualized the results using the multimodal activation map in Eq. 3 (see detailed examples in Supp. B). The results demonstrate that TAM effectively locates objects across diverse datasets, as shown in Fig. 4. These findings indicate that TAM enhances MLLMs with localization capabilities, which could be potentially beneficial for various downstream tasks, including segmentation [32], object counting [46], anomaly detection [14], image editing [6], autonomous driving [22], and ap-

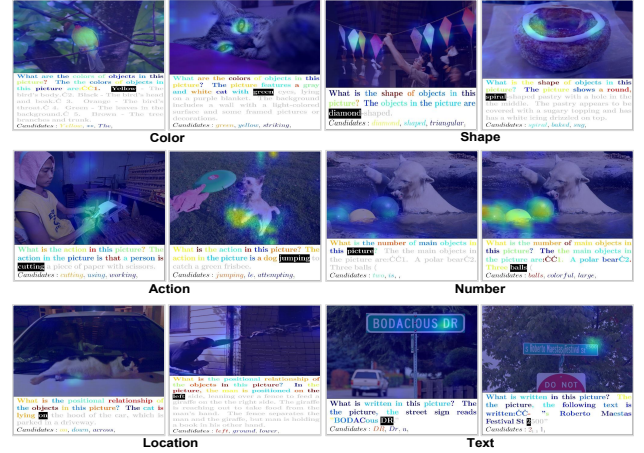


Figure 5. TAM supports to explain and analyze diverse attributes of MLLM (Qwen2-VL-7B). See extensive examples in Supp. M.

plications in the medical field [21, 50], which require pixel-level activations.

Visualization for Various Attributes. We apply TAM to explain various attributes of MLLMs, specifically focusing on fine-grained token types as Fig. 5. The results indicate that TAM effectively supports the explanation of attributes such as color, shape, action, number, location, and text. Notably, the representation of numbers is less pronounced compared to object localization, suggesting that some attributes may be encoded within related object tokens. In contrast, other attributes are more prominently featured.

Explaining Failure Cases. We apply TAM to analyze failure cases in the QK-VQA dataset [37] as Fig. 7. Our findings indicate that while the model successfully locates objects, it struggles to align them with specific knowledge. For instance, the model can identify the desert and focus on the key prompt (“year”) when outputting “don’t know.” It suggests a misalignment between the desert and the year, rather than recognition or prompt issues. Besides, some responses consist of synonyms, hypernyms, or hyponyms of the correct answers, resulting in mismatches. For more case studies, please refer to the catalog in Supp. Table 5.



Figure 6. TAM significantly improves the quality of video visualization, with much fewer redundant activation and noises on STAR [52] dataset using Qwen2-VL-2B [51]. Please see extensive examples in Supp. 27 and failure cases analysis for videos in Fig. 25.



Figure 7. TAM supports failure cases analysis on QK-VQA [37]. See more cases in Supp. about VQA (Fig. 24), videos (Fig. 25), visual reasoning (Fig. 28), and multi-turn conversation (Fig. 31).

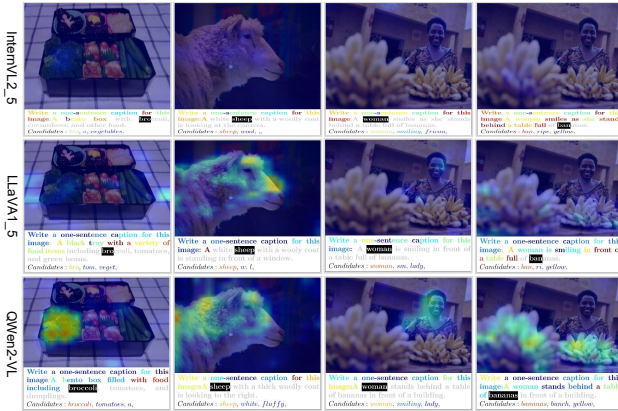


Figure 8. TAM enables qualitative comparison among MLLMs. Extensive examples are given in Fig. 17 and Fig. 18 in Supp. K.

Video Visualization. The proposed TAM excels at video visualization, where there are much fewer redundant activations and noises as shown in Fig. 6. Our method provides very clear results, while the conventional CAM makes it hard to see the video content without obvious highlights.

A Tool to Visually Comparison MLLMs. TAM can be used as a tool to compare MLLMs qualitatively. If an MLLM exhibits superior activation maps, it can be inferred that its explainability and degree of vision alignment are also better, offering valuable pixel-level predictions at no

additional cost. As shown in Fig. 8, Qwen2-VL-7B [51] shows greater explainability than LLaVA1.5-7B [35] and InternVL2.5-8B [16]. Furthermore, we observe that InternVL tends to focus more on text, resulting in weaker activation levels; in some cases, this performance is even better than that of LLaVA (e.g., with the broccoli). More extensive examples can be found in Supp. K.

5. Conclusion

This work focuses on the unique nature of MLLMs in visual explainability, which generates multiple tokens progressively, complicating the explanation process. In response, we introduced the Token Activation Map (TAM), a novel method that utilizes estimated causal inference to mitigate interferences from context tokens, with a rank Gaussian filter to reduce activation noise, thereby providing clearer visual explanations. Our results demonstrate that TAM significantly outperforms existing SoTA methods, offering high-quality visualizations applicable to diverse scenarios.

Despite the success, we focus on visual inputs, other modalities, such as audio, remain under-explored. Additionally, interpreting model decisions is an extensible further aspect. The potential applications of our TAM method are extensive, including open-vocabulary segmentation, detection, image grounding, anomaly detection, remote sensing, and medical fields that require pixel-level activations. Our work lays a foundation for advancing their explainability and practical utility across various domains.

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